Easy (10)

1.

What does NAT stand for?
A) Network Address Translation
B) Network Access Terminal
C) Network Allocation Table
D) Network Address Table
Answer: A) Network Address Translation
Explanation: NAT translates private IP addresses to public IP addresses.

2.

Which IPv6 address is used only within a local network segment?
A) Global Unicast
B) Link-local
C) Multicast
D) Broadcast
Answer: B) Link-local
Explanation: Link-local addresses (starting with FE80::) are valid only within a local subnet.

3.

What is the main reason NAT is used in IPv4 networks?
A) To enhance security by encrypting data
B) To conserve public IP addresses
C) To route multicast traffic
D) To increase bandwidth
Answer: B) To conserve public IP addresses
Explanation: NAT allows multiple devices to share one public IP.

4.

Which WAN technology uses dedicated, leased point-to-point lines?
A) MPLS
B) Leased lines
C) DSL
D) Satellite
Answer: B) Leased lines
Explanation: Leased lines provide dedicated private connections.

5.

What size is an IPv6 address?
A) 32 bits
B) 64 bits
C) 128 bits
D) 256 bits
Answer: C) 128 bits
Explanation: IPv6 addresses are 128 bits long.

6.

What protocol typically uses port numbers for NAT overloading?
A) UDP
B) ICMP
C) TCP
D) FTP
Answer: C) TCP
Explanation: PAT uses port numbers (TCP/UDP) to differentiate connections.

7.

Which IPv6 address type is used for one-to-many communication?
A) Unicast

B) Multicast
C) Anycast
D) Broadcast
Answer: B) Multicast
Explanation: Multicast sends data to multiple hosts.

8.

Which WAN technology uses virtual circuits to connect sites?
A) Frame Relay
B) DSL
C) Satellite
D) Cable Modem
Answer: A) Frame Relay
Explanation: Frame Relay uses virtual circuits over shared networks.

9.

Which feature in IPv6 replaces the IPv4 broadcast?
A) Multicast
B) Anycast
C) Link-local
D) Broadcast remains the same
Answer: A) Multicast
Explanation: IPv6 does not have broadcasts; multicast is used instead.

10.

Which IPv6 address block is reserved for Unique Local Addresses?
A) FE80::/10
B) FC00::/7
C) 2001::/16
D) FF00::/8

Answer: B) FC00::/7
Explanation: FC00::/7 is the block for Unique Local Addresses (private IPv6).

Medium (30)

11.

What type of NAT maps a private IP address to a fixed public IP address?
A) Dynamic NAT
B) Static NAT
C) PAT
D) Overloading
Answer: B) Static NAT
Explanation: Static NAT provides one-to-one mapping between private and public IPs.

12.

Which IPv6 mechanism allows devices to configure their own addresses automatically?
A) DHCPv6
B) SLAAC
C) NAT
D) ARP
Answer: B) SLAAC (Stateless Address Autoconfiguration)
Explanation: SLAAC allows devices to self-configure IPv6 addresses without DHCP.

13.

In NAT Overload (PAT), what differentiates multiple devices sharing one public IP?

A) IP addresses
B) Port numbers
C) MAC addresses
D) VLAN tags
Answer: B) Port numbers
Explanation: PAT uses different source port numbers to differentiate connections.

14.

Which WAN technology provides label switching for faster routing?
A) Frame Relay
B) MPLS
C) ATM
D) DSL
Answer: B) MPLS
Explanation: MPLS uses labels to forward packets quickly.

15.

Which IPv6 address type delivers data to the nearest device in a group?
A) Unicast
B) Multicast
C) Anycast
D) Broadcast
Answer: C) Anycast
Explanation: Anycast sends data to the nearest of multiple possible receivers.

16.

What is the IPv6 header size?
A) 20 bytes
B) 40 bytes
C) 60 bytes

D) 80 bytes
Answer: B) 40 bytes
Explanation: IPv6 header size is fixed at 40 bytes, smaller and simpler than IPv4.

17.

Which command shows the NAT translation table on a Cisco router?
A) show ip nat translations
B) show nat table
C) show ip route nat
D) show nat config
Answer: A) show ip nat translations
Explanation: This command displays current NAT mappings.

18.

Which of these is NOT a valid IPv6 address format?
A) 2001:db8::1
B) ::1
C) 192.168.1.1
D) fe80::1234
Answer: C) 192.168.1.1
Explanation: That is an IPv4 address, not IPv6.

19.

What is the primary purpose of the IPv6 extension headers?
A) Add routing information
B) Provide optional internet-layer information
C) Encrypt packet data
D) Replace the entire header
Answer: B) Provide optional internet-layer information
Explanation: Extension headers add optional info without changing the main header.

20.

What is the maximum number of devices supported by IPv4 compared to IPv6?
A) 2^32 vs 2^128
B) 2^16 vs 2^64
C) 2^64 vs 2^128
D) 2^32 vs 2^64
Answer: A) 2^32 vs 2^128
Explanation: IPv4 supports about 4 billion addresses; IPv6 supports a vastly larger number.

21.

Which WAN technology uses asynchronous transfer mode cells of fixed size?
A) Frame Relay
B) MPLS
C) ATM
D) DSL
Answer: C) ATM
Explanation: ATM uses fixed 53-byte cells for data transport.

22.

Which IPv6 address scope is used for communication within a single site?
A) Link-local
B) Global Unicast
C) Unique Local Address (ULA)
D) Multicast
Answer: C) Unique Local Address (ULA)
Explanation: ULAs are like private IPv4 addresses, used inside an organization.

23.

In NAT, which type of address is translated to a global public IP?

A) Loopback IP
B) Private IP
C) Broadcast IP
D) Multicast IP
Answer: B) Private IP
Explanation: NAT translates private IPs to public IPs for internet communication.

24.

Which IPv6 address starts with FF00::/8?
A) Multicast
B) Anycast
C) Link-local
D) Global Unicast
Answer: A) Multicast
Explanation: FF00::/8 is reserved for multicast addresses.

25.

What technology allows WAN link aggregation for better bandwidth?
A) MPLS
B) VPN
C) Link Aggregation Control Protocol (LACP)
D) DSL
Answer: C) Link Aggregation Control Protocol (LACP)
Explanation: LACP bundles multiple physical links into one logical link.

26.

Which IPv6 feature replaces ARP in IPv4?
A) SLAAC
B) Neighbor Discovery Protocol (NDP)
C) DHCPv6

D) DNS
Answer: B) Neighbor Discovery Protocol (NDP)
Explanation: NDP manages address resolution and discovery in IPv6.

27.

What is the default TTL value in IPv6 packets?
A) 64
B) 128
C) 255
D) There is no TTL; it uses Hop Limit
Answer: D) There is no TTL; it uses Hop Limit
Explanation: IPv6 replaces TTL with Hop Limit for packet lifetime.

28.

Which of the following is NOT a WAN access technology?
A) DSL
B) Satellite
C) Ethernet LAN
D) Cable modem
Answer: C) Ethernet LAN
Explanation: Ethernet is a LAN technology, not WAN.

29.

How does MPLS improve routing efficiency?
A) By encapsulating data in IPsec
B) By using short labels to forward packets
C) By using NAT translations
D) By encrypting traffic
Answer: B) By using short labels to forward packets
Explanation: MPLS uses labels to reduce routing table lookups.

30.

In IPv6, how many bits are used for the network prefix by default?
A) 32
B) 48
C) 64
D) 128
Answer: C) 64
Explanation: The standard subnet prefix length is 64 bits for network.

Hard (10)

31.

What happens to an IPv6 packet if the Hop Limit reaches zero?
A) Packet is discarded and an ICMPv6 Time Exceeded message is sent
B) Packet is forwarded anyway
C) Packet is rerouted
D) Packet header is reset
Answer: A) Packet is discarded and an ICMPv6 Time Exceeded message is sent
Explanation: IPv6 routers discard packets with zero Hop Limit and notify the sender.

32.

Which NAT type maintains a pool of public addresses and assigns them dynamically?
A) Static NAT
B) Dynamic NAT
C) PAT

D) Overload NAT
Answer: B) Dynamic NAT
Explanation: Dynamic NAT assigns public IPs from a pool dynamically.

33.

Which IPv6 extension header provides routing information similar to source routing?
A) Hop-by-Hop Options
B) Routing header
C) Fragment header
D) Destination Options
Answer: B) Routing header
Explanation: Routing header specifies intermediate nodes for packet forwarding.

34.

Which WAN technology uses labels to switch packets in the provider network?
A) Frame Relay
B) MPLS
C) ATM
D) DSL
Answer: B) MPLS
Explanation: MPLS uses label switching inside the provider's network.

35.

In IPv6, how is fragmentation handled differently than IPv4?
A) Routers fragment packets
B) Only the source node fragments packets
C) IPv6 does not support fragmentation
D) Fragmentation is done at the data link layer
Answer: B) Only the source node fragments packets
Explanation: In IPv6, only the sending host fragments packets, routers do not.

36.

Which NAT configuration allows multiple private IPs to use a single public IP with unique ports?
A) Static NAT
B) Dynamic NAT
C) PAT (NAT Overload)
D) Hybrid NAT
Answer: C) PAT (NAT Overload)
Explanation: PAT uses ports to multiplex multiple internal hosts over one IP.

37.

Which IPv6 address type is designed to be assigned to multiple devices for load balancing?
A) Unicast
B) Multicast
C) Anycast
D) Broadcast
Answer: C) Anycast
Explanation: Anycast addresses allow routing to the nearest device in a group.

38.

What is the primary function of the IPv6 Neighbor Discovery Protocol?
A) Resolve IP to MAC addresses
B) Assign IP addresses
C) Provide encryption
D) Forward packets
Answer: A) Resolve IP to MAC addresses
Explanation: NDP replaces ARP to discover neighbors and maintain reachability.

39.

Which WAN technology typically offers the lowest latency?

A) Satellite
B) Leased line
C) DSL
D) Cable modem
Answer: B) Leased line
Explanation: Leased lines provide dedicated circuits with minimal delay.

40.

What is the address format of an IPv6 Unique Local Address (ULA)?
A) FC00::/7
B) FE80::/10
C) FF00::/8
D) 2000::/3
Answer: A) FC00::/7
Explanation: ULAs use the FC00::/7 block for private IPv6 addresses.